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CONGRUENCES FOR SUMS OF BINOMIAL COEFFICIENTS

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Abstract. Let $m > 0$ and $q > 1$ be relatively prime integers. We find an explicit period $\nu_m(q)$ such that for any integers $n \geqslant 0$ and r we have

$$\begin{bmatrix} n+\nu_m(q) \\ r \end{bmatrix}_m (a) \equiv \begin{bmatrix} n \\ r \end{bmatrix}_m (a) \pmod{q},$$

provided that $a = -1$ and $n \neq 0$, or a is an integer with $1-(-a)^m$ relatively prime to q, where $\left[{n \atop r}\right]_m(a) = \sum_{k\equiv r \pmod{m}} \binom{n}{k} a^k$. This is a further extension of a congruence of Glaisher.

1. Introduction

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$. Following [S95, S02], for $m \in \mathbb{Z}^+$, $n \in \mathbb{N}$ and $r \in \mathbb{Z}$ we set

$$\begin{bmatrix} n \\ r \end{bmatrix}_m = \sum_{\substack{0\leqslant k\leqslant n \\ k\equiv r \pmod{m}}} \binom{n}{k} = |\{X \subseteq \{1,\ldots,n\} : |X| \equiv r \pmod{m}\}|. \tag{1.1}$$

Such sums occur in several topics of number theory or combinatorics. (See, e.g., [SS, H, GS, S02].)

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Let p be an odd prime. In 1899 J. W. L. Glaisher obtained the following congruence:

$$\begin{bmatrix} n+p-1 \\ r \end{bmatrix}_{p-1} \equiv \begin{bmatrix} n \\ r \end{bmatrix}_{p-1} \pmod p \quad \text{for any } n \in \mathbb{Z}^+ \text{ and } r \in \mathbb{Z}.$$

Since an odd integer is not divisible by $p-1$, this implies Hermite's result that $\left[{n \atop 0}\right]_{p-1} \equiv 1 \pmod p$ for $n = 1, 3, 5, \ldots$ (cf. L. E. Dickson [D, p. 271]). A sophisticated proof of Glaisher's congruence can be found in A. Granville [G97], while a simple proof of this appeared in [S04].

Before stating our further extension of Glaisher's result, let us introduce some notations.

Let $m \in \mathbb{Z}^+$, $n \in \mathbb{N}$ and $r \in \mathbb{Z}$. We set

$$\begin{bmatrix} n \\ r \end{bmatrix}_m (a) = \sum_{\substack{0 \leqslant k \leqslant n \\ k \equiv r \ (\mathrm{mod}\ m)}} \binom{n}{k} a^k \quad \text{for } a \in \mathbb{Z}. \tag{1.2}$$

Obviously $\left[{n \atop r}\right]_m(1) = \left[{n \atop r}\right]_m$ and

$$\begin{bmatrix} n \\ r \end{bmatrix}_m (-1) = \begin{cases} (-1)^r \left[{n \atop r}\right]_m & \text{if } 2 \mid m, \\ (-1)^r \left\{{n \atop r}\right\}_m & \text{if } 2 \nmid m, \end{cases}$$

where

$$\begin{Bmatrix} n \\ r \end{Bmatrix}_m = \sum_{\substack{0 \leqslant k \leqslant n \\ k \equiv r \ (\mathrm{mod}\ m)}} (-1)^{\frac{k-r}{m}} \binom{n}{k} = \begin{bmatrix} n \\ r \end{bmatrix}_{2m} - \begin{bmatrix} n \\ r+m \end{bmatrix}_{2m}$$

as in [S02]. It is easy to see that

$$\begin{bmatrix} n+1 \\ r \end{bmatrix}_m (a) = \begin{bmatrix} n \\ r \end{bmatrix}_m (a) + a \begin{bmatrix} n \\ r-1 \end{bmatrix}_m (a). \tag{1.3}$$

Let $a, b \in \mathbb{Z}$ and $q, m, n \in \mathbb{Z}^+$. Clearly

$$\begin{aligned}
&(x+a)^n \equiv x^n + b \mod (q, x^m - 1) \\
\iff &\sum_{r=0}^{m-1} \sum_{\substack{0 \leqslant k < n \\ k \equiv r \ (\mathrm{mod}\ m)}} \binom{n}{k} x^k a^{n-k} \equiv b \mod (q, x^m - 1) \\
\iff &\sum_{\substack{0 \leqslant k < n \\ k \equiv r \ (\mathrm{mod}\ m)}} \binom{n}{k} a^{n-k} \equiv b \pmod q \text{ for } r = 0, \ldots, m-1 \\
\iff &\sum_{\substack{1 \leqslant k \leqslant n \\ k \equiv r \ (\mathrm{mod}\ m)}} \binom{n}{k} a^k \equiv b \pmod q \text{ for every integer } r.
\end{aligned}$$

(See also [G05] and [Sc].) Since the congruence condition $(x+a)^n \equiv x^n + a \bmod (n, x^m - 1)$ plays a central role in the recent polynomial time primality test given by Agrawal, Kayal and Saxena [AKS], it is interesting to investigate periodicity of $\left[{n \atop r}\right]_m(a) \bmod q$ (where $r \in \mathbb{Z}$) with respect to n.

Let $q > 1$ be an integer relatively prime to $m \in \mathbb{Z}^+$. Write q in the factorization form $\prod_{s=1}^t p_s^{\alpha_s}$ where $p_1, \dots, p_t$ are distinct primes and $\alpha_1, \dots, \alpha_t \in \mathbb{Z}^+$. We define

$$\nu_m(q) = \mathrm{lcm}[p_1^{\alpha_1 - 1}(p_1^{\beta_1} - 1), \dots, p_t^{\alpha_t - 1}(p_t^{\beta_t} - 1)], \tag{1.4}$$

where $\mathrm{lcm}[n_1, \dots, n_t]$ represents the least common multiple of those $n_s \in \mathbb{Z}^+$ with $1 \leqslant s \leqslant t$, and each β_s is the order of p_s modulo m (i.e., β_s is the smallest positive integer with $p_s^{\beta_s} \equiv 1 \pmod{m}$). Clearly $\nu_1(q) = \mathrm{lcm}[\varphi(p_1^{\alpha_1}), \dots, \varphi(p_t^{\alpha_t})]$ divides $\varphi(q)$, where φ is Euler's totient function. Since $\varphi(p_s^{\alpha_s}) \mid \nu_m(q)$ for each $s = 1, \dots, t$, if $a \in \mathbb{Z}$ is relatively prime to q, then by Euler's theorem $a^{\nu_m(q)} \equiv 1 \pmod{p_s^{\alpha_s}}$ and therefore $a^{\nu_m(q)} \equiv 1 \pmod{q}$. Note also that $\nu_{p-1}(p^\alpha) = \varphi(p^\alpha)$ for any prime p and $\alpha \in \mathbb{Z}^+$.

Now we present our main result whose proof will be given in the next section.

Theorem 1.1. *Let $q > 1$ be an integer relatively prime to both $m \in \mathbb{Z}^+$ and $\sum_{j=0}^{m-1}(-a)^j$ where $a \in \mathbb{Z}$. Then, for any $T \in \mathbb{Z}^+$ divisible by $\nu_m(q)$, we have*

$$\sum_{k=0}^{n}(-1)^k \binom{n}{k} \left[{kT + l \atop r}\right]_m (a) \equiv \frac{(a+1)^l}{m}\left(1 - (a+1)^T\right)^n \pmod{q^n} \tag{1.5}$$

for every $n = 1, 2, 3, \dots$.

Remark 1.1. In the case $a = -1$, the right hand side of the congruence (1.5) turns out to be $\delta_{l,0}/m$ where $\delta_{l,0}$ takes 1 or 0 according as $l = 0$ or not. When $\gcd(a+1, q) = 1$ (where $\gcd(a+1, q)$ denotes the greatest common divisor of $a+1$ and q), $1 - (a+1)^T$ is a multiple of q since $\nu_m(q) \mid T$.

Corollary 1.1. *Let $a, r \in \mathbb{Z}$, $l \in \mathbb{N}$ and $m \in \mathbb{Z}^+$. Let $q > 1$ be an integer relatively prime to $m\sum_{j=0}^{m-1}(-a)^j$. Then*

$$\left[{l + \nu_m(q) \atop r}\right]_m (a) - \left[{l \atop r}\right]_m (a) \equiv \begin{cases} 0 \pmod{q_0}, \\ -(a+1)^l/m \pmod{q/q_0}, \end{cases} \tag{1.6}$$

where q_0 is the largest divisor of q relatively prime to $a+1$. Moreover, for *any $k, n \in \mathbb{Z}^+$ we have*

$$\begin{aligned}&\begin{bmatrix}k\nu_m(q)+l\\ r\end{bmatrix}_m(a)-\frac{(a+1)^l}{m}\sum_{n\leqslant j\leqslant k}\binom{k}{j}\left((a+1)^{\nu_m(q)}-1\right)^j\\ \equiv&\sum_{j=0}^{n-1}(-1)^{n-1-j}\binom{k-1-j}{n-1-j}\binom{k}{j}\begin{bmatrix}j\nu_m(q)+l\\ r\end{bmatrix}_m(a)\ (\mathrm{mod}\ q^n).\end{aligned}\tag{1.7}$$

Proof. Suppose that $p^\alpha \| q$ (i.e., $p^\alpha \mid q$ but $p^{\alpha+1} \nmid q$) where p is a prime and $\alpha \in \mathbb{Z}^+$. If $p \nmid a+1$, then $(a+1)^{\nu_m(q)} - 1 \equiv 0 \pmod{p^\alpha}$ by Euler's theorem; if $p \mid a+1$, then $p^\alpha | (a+1)^{\nu_m(q)}$ since $\nu_m(q) \geqslant p^{\alpha-1} \geqslant \alpha$. Thus (1.6) follows from (1.5) in the case $n = 1$ and $T = \nu_m(q)$.

Let $k, n \in \mathbb{Z}^+$. By Lemma 2.1 of [Su],

$$a_k=\sum_{j=0}^{n-1}(-1)^{n-1-j}\binom{k-1-j}{n-1-j}\binom{k}{j}a_j+\sum_{n\leqslant j\leqslant k}\binom{k}{j}(-1)^j\sum_{i=0}^{j}\binom{j}{i}(-1)^ia_i$$

for any sequence $a_0, a_1, \dots$ of numbers. Applying this we immediately obtain (1.7) from Theorem 1.1. $\square$

Remark 1.2. (i) When q is a prime, $m = q-1$ and $a = -1$, the first part of Corollary 1.1 yields the result of Glaisher. Previous proofs of Glaisher's congruence do not work for our generalization. (ii) Let $k, m \in \mathbb{Z}^+$, and let $q > 1$ be an integer with $\gcd(m, q) = 1$. Putting $a = -1$ and $n = 2$ in (1.7) we obtain that

$$\begin{aligned}&\begin{bmatrix}k\nu_m(q)+l\\ r\end{bmatrix}_m(-1)-k\begin{bmatrix}\nu_m(q)+l\\ r\end{bmatrix}_m(-1)\\ \equiv&-(k-1)\left(\begin{bmatrix}l\\ r\end{bmatrix}_m(-1)-\frac{\delta_{l,0}}{m}\right)\pmod{q^2}\end{aligned}$$

for any $l \in \mathbb{N}$ and $r \in \mathbb{Z}$. In particular, if $m = p-1$ and $q = p$ where p is an odd prime, then

$$\begin{bmatrix}k(p-1)\\ r\end{bmatrix}_{p-1}\equiv k\binom{p-1}{r}-(-1)^r(k-1)(p+1)+\delta_{r,0}\ (\mathrm{mod}\ p^2)\tag{1.8}$$

for every $r = 0, 1, \dots, p-2$. (1.8) in the case $r = 0$ solves a problem proposed by V. Dimitrov [Di].

Let p be a prime and let $\alpha, n \in \mathbb{Z}^+$. By Corollary 1.1, we have

$$\begin{bmatrix}p^\alpha n\\ r\end{bmatrix}_{p-1}\equiv\begin{bmatrix}p^{\alpha-1}n\\ r\end{bmatrix}_{p-1}\ (\mathrm{mod}\ p^\alpha)\ \text{ for any } r\in\mathbb{Z}.\tag{1.9}$$

In 1953, by using some deep properties of Bernoulli numbers, L. Carlitz [C] extended Hermite's congruence in the following way:

$$p+(p-1)\sum_{\substack{0<k<p^{\alpha-1}n\\ p-1\mid k}}\binom{p^{\alpha-1}n}{k}\equiv 0 \pmod{p^{\alpha}}$$

provided that p is an odd prime. When $p-1 \mid n$, this follows from Corollary 1.1, for, $\nu_{p-1}(p^{\alpha})$ divides $p^{\alpha-1}n$ and hence

$$\begin{bmatrix}p^{\alpha-1}n\\ 0\end{bmatrix}_{p-1}(-1)\equiv\begin{bmatrix}0\\ 0\end{bmatrix}_{p-1}(-1)-\frac{1}{p-1}=1-\frac{1}{p-1} \pmod{p^{\alpha}}.$$

Let $m>0$ and $q>1$ be relatively prime integers. Let a be an integer with $a\equiv -1 \pmod q$ or $\gcd(1-(-a)^m,q)=1$. What is the smallest positive integer $\mu_m(a,q)$ such that

$$\begin{bmatrix}n+\mu_m(a,q)\\ r\end{bmatrix}_m(a)\equiv\begin{bmatrix}n\\ r\end{bmatrix}_m(a) \pmod q \tag{1.10}$$

holds for all $n\in\mathbb{Z}^+$ and $r\in\mathbb{Z}$? Clearly $\mu_m(0,q)=1$, and $\mu_m(a,q)|\nu_m(q)$ by Corollary 1.1. (If $\mu_m(a,q)\nmid\nu_m(q)$, then the least positive residue of $\nu_m(q)$ mod $\mu_m(a,q)$ would be a period less than $\mu_m(a,q)$.)

Conjecture 1.1. *Let $m>0$ and $q>1$ be relatively prime integers with $q\not\equiv 0 \pmod 3$. Then $\nu_m(q)$ is the maximal value of $\mu_m(a,q)$ where a is an integer with $a\equiv -1 \pmod q$ or $\gcd(1-(-a)^m,q)=1$.*

Now we give an example to illustrate our conjecture.

Example 1.1. (i) Since the order of 3 modulo 7 is 6, we have $\nu_7(9)=3(3^6-1)=2184$. For any given $a\in\mathbb{Z}$, clearly

$$1-(-a)^7=1+a^3a^3a\equiv 1+a^3\equiv 1+a \pmod 3$$

since $a^3\equiv a \pmod 3$, thus $\gcd(1-(-a)^7,9)=1$ if and only if $a\not\equiv 2 \pmod 3$. Through computation we obtain that

$$\mu_7(-1,9)=1092,\ \mu_7(1,9)=\mu_7(-2,9)=\mu_7(4,9)=546,\ \mu_7(\pm 3,9)=3.$$

(ii) The order of 5 modulo 7 is 6, thus $\nu_7(5)=5^6-1=15624$. For any given $a\in\mathbb{Z}$, clearly $1-(-a)^7=1+a^5a^2\equiv 1+a^3 \pmod 5$, thus $5\nmid 1-(-a)^7$ if and only if $a\not\equiv -1 \pmod 5$. By computation we find that

$$\mu_7(1,5)=868,\ \mu_7(-1,5)=1736,\ \mu_7(2,5)=2232,\ \mu_7(-2,5)=15624.$$

2. Proof of Theorem 1.1

In this section we work with congruences in the ring of algebraic integers. The reader may consult §1 of [IR, Chapter 6] for the basic knowledge of algebraic integers.

Lemma 2.1. *Let $a \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, and let $q > 1$ be an integer relatively prime to $m\sum_{j=0}^{m-1}(-a)^j$. If $\zeta \neq 1$ is an m-th root of unity, then we have the congruence*

$$(1+a\zeta)^{\nu_m(q)} \equiv 1 \pmod q \tag{2.1}$$

in the ring of algebraic integers.

Proof. Let p be any prime divisor of q, and let β be the order of p modulo m. Below we use induction to show that

$$(1+a\zeta)^{p^{\alpha-1}(p^\beta-1)} \equiv 1 \pmod{p^\alpha} \tag{2.2}$$

for every $\alpha = 1, 2, 3, \ldots$.

Since $p \mid \binom{p}{k}$ for $k = 1, \ldots, p-1$ and $a^p \equiv a \pmod p$ by Fermat's little theorem, we have

$$(1+a\zeta)^p = 1 + a^p\zeta^p + \sum_{k=1}^{p-1}\binom{p}{k}(a\zeta)^k \equiv 1 + a\zeta^p \pmod p,$$

hence

$$(1+a\zeta)^{p^2} \equiv (1+a\zeta^p)^p \equiv 1 + a\zeta^{p^2} \pmod p$$

and so on. Thus

$$(1+a\zeta)^{p^\beta} \equiv 1 + a\zeta^{p^\beta} = 1 + a\zeta \pmod p.$$

(Recall that $p^\beta \equiv 1 \pmod m$ and $\zeta^m = 1$.) Clearly

$$\begin{aligned}\prod_{0<j<m}\frac{1+ae^{2\pi ij/m}}{-e^{2\pi ij/m}} &= \prod_{0<j<m}(x - e^{-2\pi ij/m})\Big|_{x=-a} \\ &= \lim_{x\to -a}\frac{x^m-1}{x-1} = \sum_{j=0}^{m-1}(-a)^j\end{aligned}$$

and so $z = 1 + a\zeta$ divides $c = \sum_{j=0}^{m-1}(-a)^j$ in the ring of algebraic integers. Therefore

$$cz^{p^\beta-1} \equiv \frac{c}{z}z^{p^\beta} \equiv \frac{c}{z}z \equiv c \pmod p$$

and hence $z^{p^\beta-1} \equiv 1 \pmod p$ since $p \nmid c$. This proves (2.2) in the case $\alpha = 1$.

Now let $\alpha \in \mathbb{Z}^+$ and suppose that (2.2) holds. Then $z^{p^{\alpha-1}(p^\beta-1)} = 1 + p^\alpha\omega$ for some algebraic integer ω. It follows that

$$z^{p^\alpha(p^\beta-1)} = (1+p^\alpha\omega)^p \equiv 1 + \binom{p}{1}p^\alpha\omega \equiv 1 \pmod{p^{\alpha+1}}.$$

This concludes the induction step.

For any $q_1, q_2 \in \mathbb{Z}$ with $\gcd(q_1, q_2) = 1$, there are $x_1, x_2 \in \mathbb{Z}$ such that $q_1x_1 + q_2x_2 = 1$, If an algebraic integer ω is divisible by both q_1 and q_2, then $\omega = q_1(\omega x_1) + q_2(\omega x_2)$ is divisible by q_1q_2 in the ring of algebraic integers. Therefore (2.1) is valid in view of what we have proved. □

Remark 2.1. Write an integer $q > 1$ in the form $p_1^{\alpha_1} \cdots p_t^{\alpha_t}$, where $p_1, \ldots, p_t$ are distinct primes and $\alpha_1, \ldots, \alpha_t \in \mathbb{Z}^+$. Let m be a positive integer dividing $p_s - 1$ for all $s = 1, \ldots, t$. And let g be an integer with $g \equiv g_s^{\varphi(p_s^{\alpha_s})/m} \pmod{p_s^{\alpha_s}}$ for $s = 1, \ldots, t$, where g_s is a primitive root modulo p_s. Clearly $g^m \equiv 1 \pmod q$. Suppose that $j \in \mathbb{Z}^+$ and $j < m$. Then $p_s - 1 \nmid j\varphi(p_s^{\alpha_s})/m$ and hence $g^j \not\equiv 1 \pmod{p_s}$. Therefore $\gcd(g^j - 1, q) = 1$. If $p_s \mid 1 + ag^j$, then $-a \equiv g^{m-j} \not\equiv 1 \pmod{p_s}$ but $(a+1)\sum_{i=0}^{m-1}(-a)^i = 1 - (-a)^m \equiv 1 - (-ag^j)^m \equiv 0 \pmod{p_s}$. Thus, if $\gcd(\sum_{i=0}^{m-1}(-a)^i, q) = 1$, then $\gcd(1 + ag^j, q) = 1$, and hence

$$(1 + ag^j)^{\nu_m(q)} \equiv 1 \pmod q \tag{2.3}$$

which is an analogue of (2.1).

Proof of Theorem 1.1. Set $\zeta = e^{2\pi i/m}$. For any $l \in \mathbb{Z}$, we clearly have

$$\sum_{j=0}^{m-1} \zeta^{jl} = \begin{cases} m & \text{if } m \mid l, \\ 0 & \text{otherwise.} \end{cases}$$

If $n \in \mathbb{N}$ then

$$\begin{aligned} m\begin{bmatrix} n \\ r \end{bmatrix}_m(a) &= \sum_{k=0}^{n}\binom{n}{k}a^k\sum_{j=0}^{m-1}\zeta^{j(k-r)} \\ &= \sum_{j=0}^{m-1}\zeta^{-jr}\sum_{k=0}^{n}\binom{n}{k}a^k\zeta^{jk} = \sum_{j=0}^{m-1}\zeta^{-jr}(1+a\zeta^j)^n. \end{aligned}$$

Now let $T \in \mathbb{Z}^+$ be a multiple of $\nu_m(q)$, and fix a positive integer n. By the above,

$$\begin{aligned}
&m\sum_{k=0}^{n}(-1)^k\binom{n}{k}\begin{bmatrix}kT+l\\ r\end{bmatrix}_m(a)\\
=&\sum_{j=0}^{m-1}\zeta^{-jr}\sum_{k=0}^{n}\binom{n}{k}(-1)^k(1+a\zeta^j)^{kT+l}\\
=&\sum_{j=0}^{m-1}\zeta^{-jr}(1+a\zeta^j)^l\left(1-(1+a\zeta^j)^T\right)^n\\
\equiv&(1+a)^l\left(1-(1+a)^T\right)^n \pmod{q^n}
\end{aligned}$$

where we apply Lemma 2.1 in the last step. This concludes our proof. $\square$

Remark 2.2. Let $a,r\in\mathbb{Z}$ and $m\in\mathbb{Z}^+$, and let $q>1$ be an integer relatively prime to $\sum_{j=0}^{m-1}(-a)^j$. Suppose that $m\mid p-1$ for any prime divisor p of q. Obviously $\gcd(m,q)=1$. Choose $g\in\mathbb{Z}$ as in Remark 2.1. Then $g^m\equiv 1\pmod q$, and for each $0<j<m$ we have $\gcd(g^j-1,q)=1$ as well as (2.3). By modifying the proof of Theorem 1.1 slightly, we find that

$$m\begin{bmatrix}n\\ r\end{bmatrix}_m(a)\equiv\sum_{j=0}^{m-1}g^{-jr}(1+ag^j)^n=(a+1)^n+\sum_{0<j<m}g^{-jr}a_j^n \pmod q$$

for every $n\in\mathbb{N}$, where $a_j=1+ag^j$ $(0<j<m)$ are relatively prime to q. If $q\mid a+1$ or $\gcd(a+1,q)=1$, then the function $f:\mathbb{Z}^+\to\mathbb{Z}$ given by $f(n)=\left[\begin{smallmatrix}n\\ r\end{smallmatrix}\right]_m(a)$ is q-normal in the sense that

$$f(n)\equiv\sum_{\substack{1\leqslant j<q\\ \gcd(j,q)=1}}c_jj^n \pmod q\qquad\text{for all } n\in\mathbb{Z}^+,\tag{2.4}$$

where c_j $(1\leqslant j<q$ and $\gcd(j,q)=1)$ are suitable integers. The concept of q-normal function was first introduced by Sun [S03] where the reader can find some q-normal functions involving Bernoulli polynomials.

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